

The Surgeon Who Built a Robot to Democratize Heart Surgery

Dr. Sudhir Srivastava, one of the early pioneers of robotic cardiac surgery, while living in the United States, performed more robotic cardiac surgeries than anyone else in the world. Now, he's ensuring millions more can access the same life-saving technology.

Dr. Sudhir Srivastava

Founder, and Chairman
SS Innovations International Inc.



At SS Innovations' International Facility in Gurugram, Founder, Chairman, and CEO Dr. Sudhir Srivastava is leading a visible revolution in healthcare that unites surgical insight with engineering to expand access to high-precision care. The globally renowned cardiac surgeon, credited with the world's first quadruple-vessel beating-heart TECAB, has traded daily scrubs for a builder's mandate, yet the mission is unchanged: to save more lives through innovation made accessible.

Srivastava's transformation from operating-room pioneer to medical-device entrepreneur began after he introduced robotic surgery programs to India in 2011 and confronted a hard truth. Transformative technology was priced beyond reach for most hospitals and patients. Imported robotic systems cost millions, with upkeep adding to the total cost of ownership, making broad adoption impossible in a country of over a billion people.

Refusing that trade-off, he invested personal savings, borrowed money, and sold assets to build a world-class alternative. The result was the SSI Mantra, India's first indigenously developed multi-arm surgical robotic platform, engineered to rival incumbent systems while dramatically reducing costs. Designed for modularity, immersive 3D visualization, and

ergonomic control, the Mantra system enables up to five robotic arms and full-team 3D vision access, empowering hospitals that could never justify traditional robotic investments.

"The goal is not only to make the system affordable but also to address the shortage of robotics systems in countries like India, where the demand is substantial compared to availability," says Srivastava.

What sets him apart is the precision of a technologist combined with the empathy of a clinician. His focus on minimally invasive cardiac surgery grew from witnessing the grueling recoveries from open procedures. His first robotic cardiac patient left the hospital within 23 hours and drove 800 miles a week later, proving that smaller incisions and steadier hands could restore life faster.

Now serving as what he calls the "Chief Everything Officer" at SS Innovations International, Srivastava balances clinical insight, engineering leadership, and operations from the company's International Facility in Gurugram. Under his guidance, SS Innovations has evolved from early pilots to a multi-country footprint, with SSI Mantra systems installed across India and eight other countries, including Nepal, Ecuador, Guatemala, Colombia, the Philippines, Indonesia, Iraq and Sri Lanka.

Regulatory approvals are underway, with a 510K submission to the U.S. FDA in process and CE Mark certification targeted for late 2025 to early 2026.

"The ultimate standard at SSI is to create technology that meets the criteria of being used on one's own family," Srivastava states, underscoring his commitment to safety and quality.

Dr. Srivastava has built a holistic ecosystem of mixed-reality training, tele-mentoring, and robust instrumentation to enable connected care and accelerate skill transfer. That vision has already redefined boundaries. In September 2025, a 16-month-old infant underwent telesurgery using the Mantra 3 system, with surgeons operating from Gurugram to Hyderabad, nearly 1,550 km away. Earlier, Indian surgeons in Strasbourg performed complex telesurgeries, including cardiac, on patients in Indore, demonstrating intercontinental resilience in connected care.

Access also means investing where care is limited. Partnering with the Sri Madhusudan Sai Institute of Medical Sciences and Research (SMSIMSR) in Chikkaballapur, Srivastava donated a Mantra system and committed the first 100 surgeries, five years of maintenance, and cross-specialty training, creating a model for capacity building in underserved regions.

In April 2025, SS Innovations' uplisting on Nasdaq signaled readiness to scale globally, adding capital discipline and visibility to challenge long-standing incumbents. With more than 115 Mantra systems sold, 5,500 surgeries completed, and 50 telesurgeries conducted, the momentum reflects both clinical trust and manufacturing maturity.

For Srivastava, the metric of success remains human, not financial: safer procedures, shorter stays, and faster returns to life. That is the promise driving the democratization of advanced surgery, from India to the world.